10.2 Calculus with Parametric Curves

Apply calculus to parametric curves to find slopes, lengths, and areas.

Tangent & Areas

A curve $x = f(t),\ y = g(t)$ is **differentiable** at t if f and g are differentiable at t.

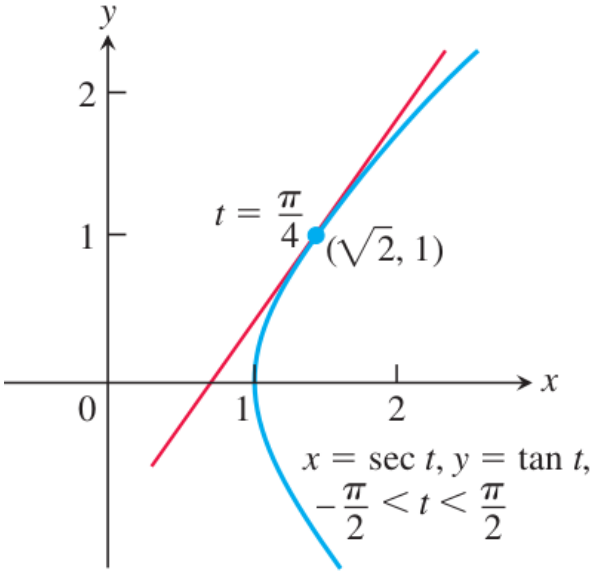
If y is a differentiable function of x, then by Chain Rule, we have

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}.$$

If y is a twice-differentiable function of x and $dx/dt \neq 0$, then writing $y' = dy/dx$, we have

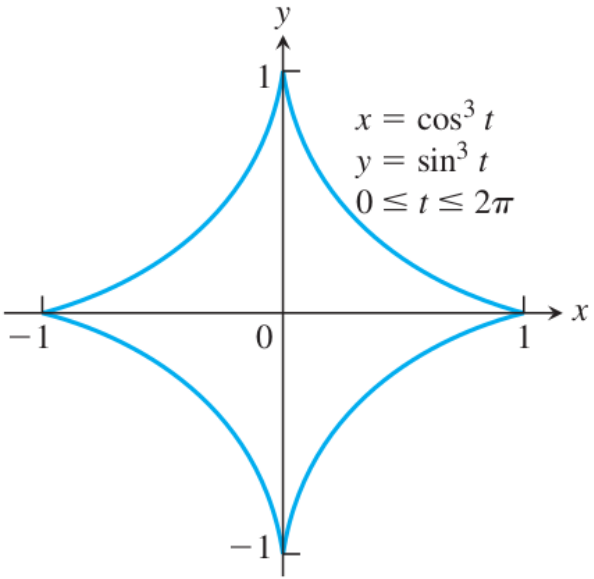
$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt}.$$

Example: Find the tangent to the curve $x = \sec t,\ y = \tan t,\ -\frac{\pi}{2} < t < \frac{\pi}{2}$, at the point $(\sqrt{2}, 1)$, where $t = \pi/4$.



Example: Find d^2y/dx^2 as a function of t if $x = t - t^2$ and $y = t - t^3$.

Example: Find the area enclosed by the astroid $x = \cos^3 t,\ y = \sin^3 t,\ 0 \le t < 2\pi$.



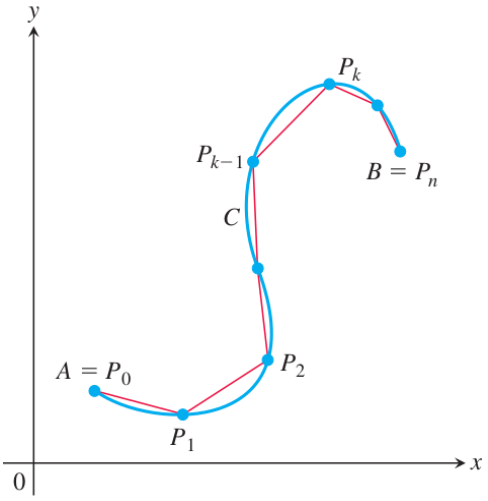
Length of a Parametrically Defined Curve

A curve C defined by $x = f(t)$ and $y = g(t)$, $a \le t \le b$ is a **smooth curve** if f and g are continuously differentiable with f' and g' not simultaneously zero.

Definition: Length of Smooth Curve

If C is a smooth curve defined by $x = f(t)$ and $y = g(t)$, $a \le t \le b$, and C is traversed exactly once at t increases from $t = a$ to $t = b$, then the **length of** C is the definite integral

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}\, dt.$$



Example: Find the length of the circle of radius r defined parametrically by

$$x = r\cos t \quad \text{and} \quad y = r\sin t, \quad 0 \leq t \leq 2\pi.$$

Example: Find the length of the astroid $x = \cos^3 t,\ y = \sin^3 t,\ 0 \leq t < 2\pi$.